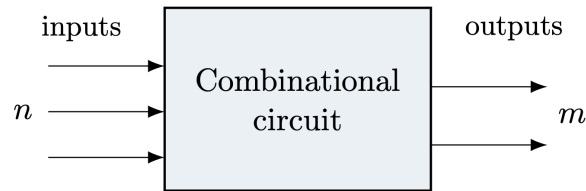


Every digital circuit we will study falls into one of two families:

1. **Combinational circuit:** built only from logic gates. Its outputs at any instant depend only on the combination of inputs present at that instant. There is no memory.
2. **Sequential circuit:** uses storage elements (memory) in addition to logic gates. Its outputs depend on the present inputs and on the stored internal state and that state depends on past inputs.

We draw a combinational circuit as a single block with n input lines coming in from outside and m output lines leaving.



Remember that there are two equivalent ways to specify the same circuit:

1. A truth table with 2^n rows, giving each output for every input combination.
2. A set of m Boolean functions, one per output, each written in terms of the n inputs.

We will tie everything that we learned in the last previous chapters for the purpose of circuit analysis. We have three goals that we want to accomplish:

1. Analyze: given a logic circuit, figure out what it does.
2. Synthesize / design: given the desired behavior, build a circuit that does it.
3. Model in HDL: write a hardware description language model of circuits.

Some combinational circuits show up so often that they get reused everywhere:

- **Adders:** add binary numbers.
- **Subtractors:** subtract binary numbers.
- **Comparators:** compare two numbers (equal, greater, less).
- **Decoders:** turn a coded input into one active output line.
- **Encoders:** turn an active line into a code.
- **Multiplexers:** select one of several inputs to pass through.

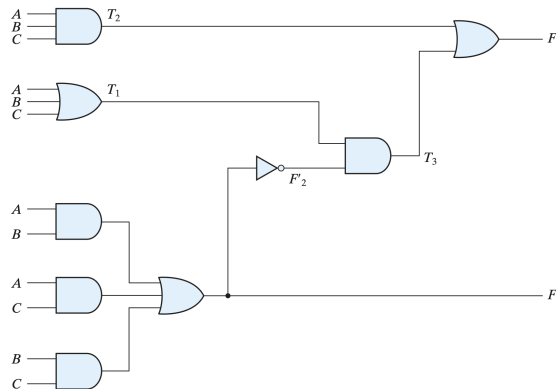
Circuit analysis: starting from a given logic diagram, determine the function the circuit implements. The end product is a set of Boolean functions, a truth table, or a clear explanation of what the circuit does. It is the reverse of design.

Steps for circuit analysis:

1. Confirm that the circuit is combinational:
 - a. A combinational circuit has no feedback paths and no memory elements. A **feedback path** is a connection from the output of one gate back to the input of an earlier gate whose output eventually feeds the first one.

2. Get the boolean function from a diagram:
 - a. Label every gate output that depends only on input variables with its own symbol (use meaningful names). Write the Boolean function for each.
 - b. Label the gates whose inputs are input variables and already-labeled gates, with new symbols. Write their Boolean functions
 - c. Repeat step 2 until you reach the circuit outputs.
 - d. By repeated substitution of the functions you defined, express each output purely in terms of the input variables.
3. Get the truth table from a diagram:
 - a. Fill in the columns for gates that depend only on the input variables.

Example: For the circuit below, conduct circuit analysis and generate a Boolean function/truth table.



Reading the diagram, the first batch of functions is:

$$F_2 = AB + AC + BC, \quad T_1 = A + B + C, \quad T_2 = ABC.$$

Next come the gates that depend on signals we just named:

$$T_3 = \overline{F_2} T_1, \quad F_1 = T_3 + T_2.$$

Now substitute back to get F_1 purely in terms of A, B, C :

$$\begin{aligned} F_1 &= T_3 + T_2 = \overline{F_2} T_1 + ABC \\ &= \overline{(AB + AC + BC)} (A + B + C) + ABC \\ &= (\overline{A} + \overline{B})(\overline{A} + \overline{C})(\overline{B} + \overline{C})(A + B + C) + ABC \\ &= (\overline{A} + \overline{B}\overline{C})(\overline{A}\overline{B} + \overline{A}\overline{C} + \overline{B}C + \overline{B}\overline{C}) + ABC \\ &= \overline{A}\overline{B}\overline{C} + \overline{A}B\overline{C} + \overline{A}\overline{B}C + ABC. \end{aligned}$$

When we solve for the boolean function of a circuit, we want to break it down into SOP form. The resulting truth table of our function is:

A	B	C	F_2	$\overline{F_2}$	T_1	T_2	T_3	F_1
0	0	0	0	1	0	0	0	0
0	0	1	0	1	1	0	1	1
0	1	0	0	1	1	0	1	1
0	1	1	1	0	1	0	0	0
1	0	0	0	1	1	0	1	1
1	0	1	1	0	1	0	0	0
1	1	0	1	0	1	0	0	0
1	1	1	1	0	1	1	0	1